

Microeconomics **Midterm 1**, Spring 2022

Name and Student ID:

1. (10 points) (5 points each) Sparty M. Spartacus is a soybean farmer with a 60-acre tract of land. Each acre can produce 100 bushels of soybeans. The costs associated with planting the tract in May total \$30,000, while the costs associated with harvesting and bringing the crop to market in September total \$10,000.
 - a. Suppose that in May, Sparty receives a forecast from an economist that says he will be able to sell his soybeans September for \$10 per bushel. Assuming the forecast is correct, should Sparty plant his crop?
 - b. Suppose Sparty does decide to plant his crop in May, but the economist fails to foresee a glut in soybean supply that causes the earlier forecast to be quite wrong. In September, but before the harvest, Sparty finds out that he will only be able to sell his soybeans for \$2 per bushel. What should Sparty do?

2. (10 points) (5 points each) (Short Questions)

- a. If a firm is producing where $MP_L/w > MP_K/r$, what can it do to reduce costs but maintain the same output?
- b. In the short run the demand for cigarettes is totally inelastic. In the long run, suppose that it is perfectly elastic. What is the impact of a specific cigarette tax on the price that consumers pay in the short run and in the long run?

3. (15 points) Suppose that a firm's fixed production function is given by

$$q = \min(5k, 10l)$$

- a. Calculate the firm's expansion path and long-run total cost functions. (10 points)

- b. Suppose that k is fixed at 10 in the short run. Calculate the firm's short-run total, and average cost functions. (5 points)

4. (20 points) Suppose that a firm's production function is given by

$$Q = \sqrt{L} + \sqrt{K} + \sqrt{M}.$$

The input prices of labor, capital, and materials are $w = 1$, $r = 1$, and $m = 1$, respectively.

- a. Given that the firm wants to produce Q units of output, what is the solution to the firm's long-run cost-minimization problem? (5 points)

- b. Given that the firm wants to produce Q units of output, what is the solution to the firm's short-run cost-minimization problem when $K = 4$? Will the firm want to use positive quantities of labor and materials for all levels of Q ? (8 points)

- c. Given that the firm wants to produce 12 units of output, what is the solution to the firm's short-run cost-minimization problem when $K = 4$ and $L = 9$? Will the firm want to use a positive quantity of materials for all levels of Q ? (7 points)

5. (35 points) (5 points each) A firm's short-run total variable cost function is given by $VC = 12Q + Q^2$, where Q is the quantity of output. The firm has chosen to utilize 9 units of capital, which cannot be adjusted in the short run no matter what quantity is produced. Assume that the rental rate of capital is \$25/unit.

a. What are the firm's fixed costs?

b. Write down the equation for the total cost function and derive the equation for average total cost.

c. What is the firm's marginal cost function?

d. Suppose the market price of the good is \$32. What is the profit-maximizing quantity, Q^* , for the firm to produce?

e. What is the firm's profit if it produces Q^* ? Would you advise the firm to produce or shut down in the short run?

f. For parts f and g, suppose there are 100 firms identical to this one in a market characterized by perfect competition. At a market price of \$32, is the market in long-run equilibrium? Assuming free entry and exit, what do you expect to happen to the number of firms in the long run?

g. What would the price have to be for this market to be in long-run equilibrium?

6. (30 points) Suppose that the market demand for a product is given by $Q=A-BP$. Suppose also that the typical firm's cost function is given by $C(q)=aq+bq^2$.

a. Compute the long-run equilibrium output and price for the typical firm in the market. (6 points)

b. Calculate the equilibrium number of firms in this market as a function of all the parameters in this problem. (5 points)

c. Describe how changes in the demand parameters A and B affect the equilibrium number of firms in this market. (6 points)

d. Describe how the parameters of the typical firm's cost function affect the long-run equilibrium number of firms in this example. (6 points)

e. Discuss how a lump-sum tax of k affects the price, firm's output, and the number of firms in the long run. (7 points)